Evaluating Sex Differences in Adaptive Behavior Development in Infants at High Risk for ASD

Institute of Child Development
COLLEGE OF EDUCATION + HUMAN DEVELOPMENT

Sharer, E.[1], Constantino, J.N.[2], Botteron, K.N.[2], Estes, A.M.[3], Hazlett, H.C.[4], Schultz, R.T.[5], Piven, J.[4], & Elison, J.T.[1] and the IBIS Network.

1. University of Minnesota, 2. Washington University St. Luis, 3. University of Washington, 4. University of North Carolina, 5. Children's Hospital of Philadelphia

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Introduction

Female Protective Effect

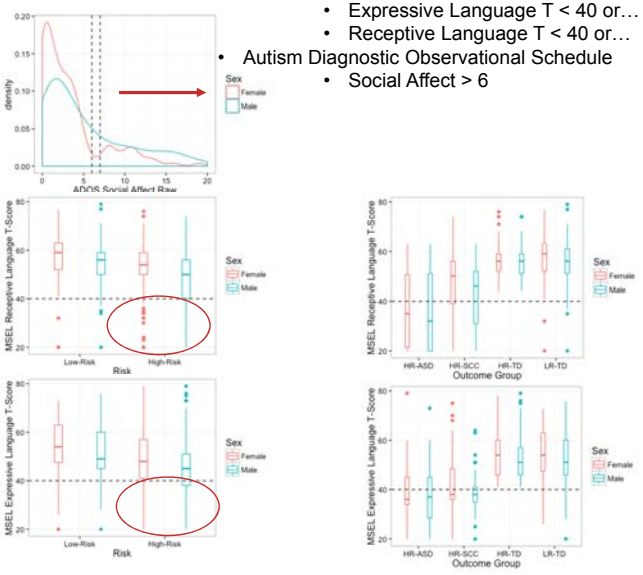
The *female protective effect* (FPE) has been hypothesized to explain the sex ratio discrepancy (i.e., 4:1) in prevalence estimates of ASD. While there is general consensus regarding the presence of an FPE, how it manifests early in development is unknown. We examined whether the FPE exerts influence in an all-or-nothing type manner as opposed to exerting influence along a gradient in 6-24 month-olds. More specifically, we compared adaptive behavior development across groups stratified by sex, diagnostic status, and concerning social-communicative behaviors. We analyzed data from 481 children recruited from the IBIS network, who contributed a total of 1665 assessments.

Does sex differentially influence behavioral trajectories in HR infant siblings with ASD, social-communicative concern, and typical outcomes?

Sample Stratification

Clinically informed outcomes were defined using 24-month data

- **Autism Spectrum Disorder (ASD)**
  - ASD/PDD-NOS DSM-IV Clinical Best Estimate Diagnoses
- Non-ASD
  - **Typical Development (TD)**
  - **Subthreshold Social-Communicative Concern (SCC)**
    - Mullen Scales of Early Learning
      - Expressive Language T < 40 or…
      - Receptive Language T < 40 or…
    - Autism Diagnostic Observational Schedule
      - Social Affect > 6



Methods

Summary Statistics and Demographics

	Female				Male			
	HR-ASD	HR-SCC	HR-TD	LR-TD	HR-ASD	HR-SCC	HR-TD	LR-TD
N	19	29	76	64	60	52	90	91
SubGroup Outcome Measures								
ADOS SA	10.11(4.21)	3.62(3.43)	1.88(1.64)	1.94(1.92)	11.88(4.05)	3.38(2.53)	1.92(1.68)	2.49(2.51)
MSEL ExpLang	40.11(13.66)	42.71(12.08)	54.4(9.86)	53.92(10.79)	37.33(12.66)	39.28(8.11)	52.51(9.39)	51.62(10.55)
MSEL RecLang	37.47(15.86)	46.82(12.43)	56.53(6.58)	57.83(9.52)	35.07(14.93)	43.32(12.49)	55.45(7)	55.34(9.04)
Completeness of Vineland Adaptive Behavioral Scales								
4 Visits	0.68	0.52	0.59	0.64	0.67	0.52	0.54	0.64
3+ Visits	0.74	0.9	0.87	0.95	0.93	0.88	0.84	0.93
2+ Visits	0.95	0.97	0.99	1	1	0.96	0.99	1
1+ Visits	1	1	1	1	1	1	1	1
Site of Data Acquisition								
PHI	3	5	21	10	20	12	19	20
SEA	8	6	13	25	13	19	18	21
STL	5	12	21	14	13	10	33	20
UNC	3	6	21	15	14	11	20	30
Maternal Education Achievement								
High School	0	3	3	0	4	5	9	2
Undergraduate	14	23	46	36	41	32	52	49
Graduate	3	2	21	28	15	11	26	37
Family Income Level								
< 50K	6	8	15	12	16	13	14	15
50K-100K	6	12	27	24	24	17	27	33
100K+	5	7	24	26	19	17	43	35

Vineland Adaptive Behavioral Scales (VABS)

Each infant had repeated parent report measures of child's adaptive level of functioning across multiple domains including communication, socialization, daily living, and motor skills.

Analytic Approach

- FPEs may be evident developmentally through:
  - Sex-specific differences in subgroup behavior
  - Distinct relative-risk ratio patterns
- Five Linear Mixed Effects (LME) models were fit to each VABS subscales and composite to capture development of adaptive behavior
  - Predictors: age, group, sex, and their interactions

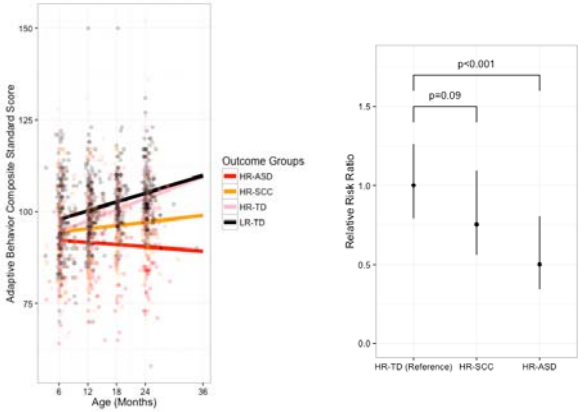
$$VABSabc_i = \beta_{00}+\beta_{10}(sex_i)+\beta_{20}(group_i)+\beta_{01}(month_i)+\beta_{11}(group_i)(month_i)+\eta_{0j}+\epsilon_i$$

  - Benjamini-Hochberg correction for pairwise comparisons
- Relative Risk (RR) Ratios comparing emergent sex ratios of the outcome groups HR-ASD and HR-SCC to HR-TD

The Infant Brain Imaging Study (IBIS) Network is an NIH funded Autism Center of Excellence (HD055741) and consists of a consortium of 8 Universities in the US and Canada.

Results

LME Models and RR Ratios



- Overall effect of sex ($p < .001$)
- No group by sex interaction ($p = .742$)
- Change rates across the first two years differentiated the groups in a consistent manner in the adaptive behavior composite, as well as the communication, and socialization domain

HR-ASD < HR-SCC < (HR-TD = LR-TD)

- Infants are less likely to be female in the HR-ASD group as compared to the HR-TD (RR=.525, p<.001) and the HR-SCC (RR=.672, p=.073). Females are trending close to being less common in the HR-SCC compared to HR-TD groups (RR=.782, p=0.087).

Conclusions and Future Directions

The sex discrepancy between in ASD is maintained in our HR sample, supporting a FPE hypothesis. There is only a trend towards a sex discrepancy for SCC, despite developmentally distinguishable differences between groups by 12 months in multiple adaptive behavior domains and entirely distinguishable across all adaptive domains by 24 months. There was no group x sex interaction.

Sex may protect from diagnostic-levels, but not subclinical-levels, of concern

Next Steps

- Compare with empirically driven stratification through growth mixture models
- Extend to subcortical and cortical measures of brain growth